

The relationship between Gross Domestic Product per capita and Life Expectancy.

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Introduction

There is a relationship between wealth and health (Pollack et al. 2007). We will explore this relationship using the relationship between [Gross Domestic Product \(GDP\) *per capita*](#) and Life Expectancy across countries and over time.

Methods

We took per capita GDP and life expectancy data from 12 years (from 1952 to 2007) for each of 142 countries. Analysis was carried out with R (R Core Team 2026) and tidyverse (Wickham et al. 2019) packages.

Results

Data about about the five most populated countries in is shown in [Table 1](#).

Table 1: Five most populated countries in 2007.

country	continent	year	lifeExp	pop	gdpPercap
China	Asia	2007	72.961	1318683096	4959.115
India	Asia	2007	64.698	1110396331	2452.210
United States	Americas	2007	78.242	301139947	42951.653
Indonesia	Asia	2007	70.650	223547000	3540.652
Brazil	Americas	2007	72.390	190010647	9065.801

Data about about the five most wealthy countries in is shown in [Table 2](#).

Table 2: Five most wealthy countries in 2007.

country	continent	year	lifeExp	pop	gdpPercap
Norway	Europe	2007	80.196	4627926	49357.19
Kuwait	Asia	2007	77.588	2505559	47306.99
Singapore	Asia	2007	79.972	4553009	47143.18
United States	Americas	2007	78.242	301139947	42951.65
Ireland	Europe	2007	78.885	4109086	40676.00

The relationship between Gross Domestic Product (GDP) per capita and Life Expectancy across countries in 1952 and 2007 is in Figure 1.

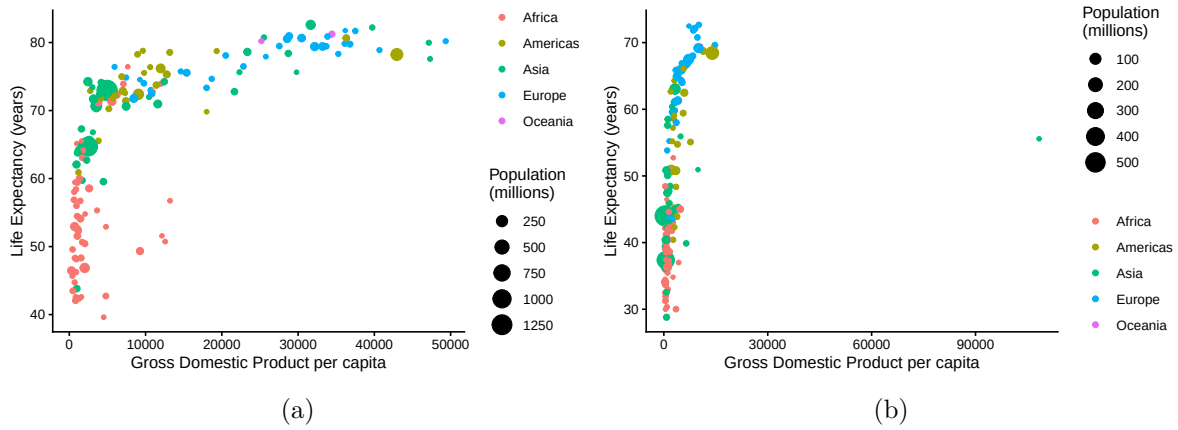


Figure 1: **The relationship between Gross Domestic Product (GDP) *per capita* and Life Expectancy across countries in 2007 (Figure 1a) and 1952 (Figure 1b).** Each point represents a country, plotted by gross domestic product (GDP) *per capita* (international dollars, x-axis) against life expectancy at birth in years (y-axis). Point colour indicates continental region and point size is proportional to population size (millions). Data are from Gapminder and were analysed with R and tidyverse packages.

Discussion

References

Pollack, Craig Evan, Sekai Chideya, Catherine Cubbin, Brie Williams, Mercedes Dekker, and Paula Braveman. 2007. "Should Health Studies Measure Wealth?: A Systematic Review."

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R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.

Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. *Welcome to the {Tidyverse}*. 4: 1686. <https://doi.org/10.21105/joss.01686>.